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with the help of Phospho Enol Pyruvic acid (PEP) and produce oxalo acetic acid (OAA). Subsequently OAA is converted into malic acid like C_4 cycle and gets accumulated in vacuole increasing the acidity. During the day time stomata are closed and malic acid is decarboxylated into pyruvic acid resulting in the decrease of acidity. CO_2 thus formed enters into Calvin Cycle and produces carbohydrates (Figure13.21).

Significance of CAM Cycle

1. It is advantageous for succulent plants to obtain CO_2 from malic acid when stomata are closed.
2. During day time stomata are closed and CO_2 is not taken but continue their photosynthesis.
3. Stomata are closed during the day time and help the plants to avoid transpiration and water loss.

13.16 Photorespiration or C_2 Cycle or Photosynthetic Carbon Oxidation (PCO) Cycle

Respiration is a continuous process for all living organisms including plants. **Decker** (1959) observed that rate of respiration is more in light than in dark. Photorespiration is the excess respiration taking place in photosynthetic cells due to absence of CO_2 and increase of O_2(Table 13.5). This

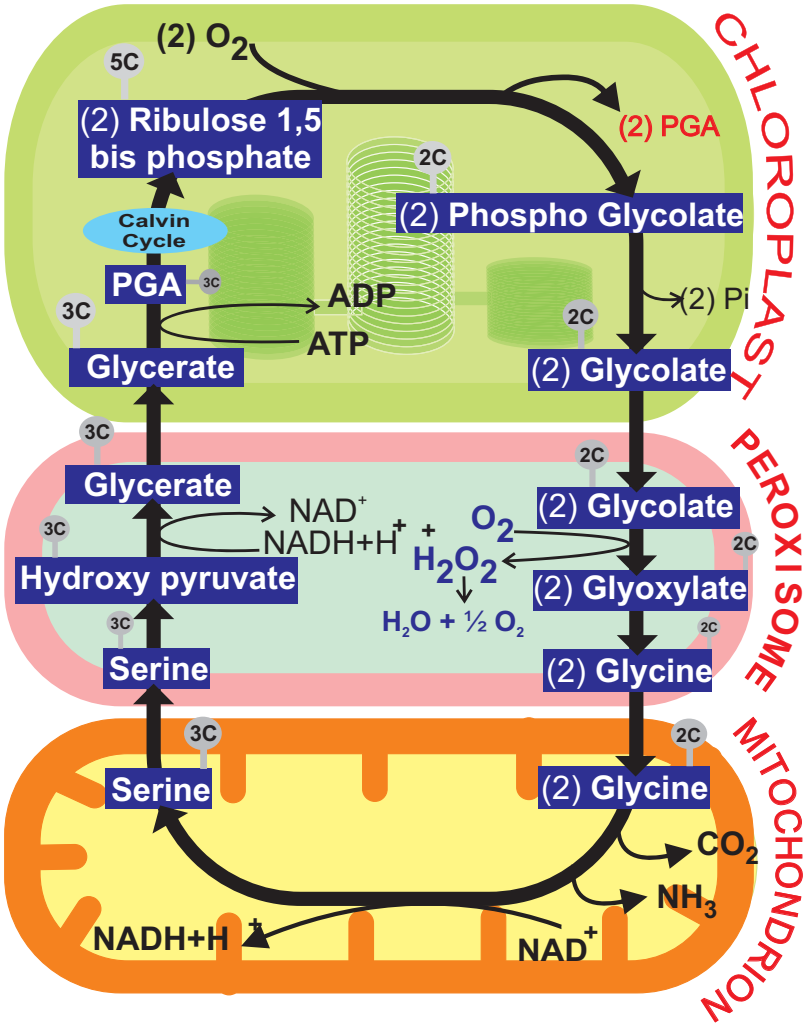


Figure 13.22: Photorespiration